

Semester Project

(CLO-5: Develop a GUI based project for a real-world problem in a team environment)

Project Overview:

Design and develop a **Smart University Campus Management System** using Java Swing that integrates academic operations, campus facilities, and student services into a unified platform. The system should simulate real-world campus activities while demonstrating advanced object-oriented programming concepts.

1. Inheritance & Abstract Classes:

- Create an abstract base class **CampusEntity** with common attributes (entityID, name, location) extend it into **AcademicUnit** (Department, Classroom, Lab), **Facility** (Library, Cafeteria, Hostel), **ServiceUnit** (TransportService, SecurityService, HealthCenter).
- Define an abstract method *calculateOperationalCost()*. Implementation examples:
 - AcademicUnit → based on number of students and equipment
 - Facility → based on maintenance and usage frequency
 - ServiceUnit → based on service hours and staff

2. Composition & Aggregation:

- A Department contains multiple Course objects & A Course contains Student and Assignment objects
- CampusZone aggregates Facilities (Library, Cafeteria) & Service units (Transport, Security). Example: A Library manages Book objects (can exist independently)

3. Interfaces:

- **Notifiable**: sendNotification() used by SecurityService, HealthCenter, Admin
- **Schedulable**: generateSchedule() used by: Course, TransportService
- **Reportable**: generateReport() used by: Department (performance), Library (usage stats)

4. GUI (Swing):

- Interactive dashboard with:

- Campus Map Panel (JPanel) to displays buildings and services and color-coded statuses (active, busy, closed)
- CRUD operation; i.e., manage students, courses, facilities using components JTextField, JComboBox etc.
- Timetable & Reports Section displays class schedules and resource usage summaries

5. File Handling: Save/load data using .dat files, with exception handling.

- All storable classes must implement Serializable
- Handle exceptions: IOException, ClassNotFoundException, Corrupted or empty file handling
- Save all system objects before closing application & Load existing data automatically on startup

6. Generics: CampusRepository<T> to manage collections of Students, Courses & Facilities

7. Polymorphism: Overridden toString() for all classes to display summaries in GUI tables.

8. Static Members: Track system-wide data like totalStudents, totalCourses, and totalFacilityUsage

Complexity Additions:

- **Interdependency Handling:**
 - If a classroom becomes unavailable → system reschedules affected classes
 - If a medical emergency occurs → notify SecurityService and HealthCenter
- **Dynamic Scheduling:**
 - TransportService adjusts routes based on peak hours
 - Course schedules update dynamically when conflicts arise
- **Data Persistence Logic:**
 - Auto-save system state periodically
 - Backup .dat files to prevent data loss
- **Role-Based Access:**

- Admin → full system access
- Teacher → manage courses and students
- Student → view schedules and reports

Deliverables:

1. **Code:** Implemented OOP concepts with GUI.
2. **UML Diagrams:** Show inheritance, composition, and interface implementations.
3. **Test Cases:** Validate where necessary/needed.
4. **Documentation:** 5-10 pages report about your system

Evaluation Enhancements:

- **Implementation of OOP Principles (25%):** Correct use of inheritance, polymorphism, interfaces, and composition.
- **Functional Completeness (20%):** All required functionalities are implemented and work as expected.
- **Documentation and Reporting (5%):** Well-documented code and clear project report.
- **Viva (50%):** Individual oral examination will be conducted.